

OSIM Lab

Phase 1 + Phase 2 Completion Report | 2026-04-02

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Author	Alec Sanchez	Phase 1	Network ✓	Phase 2	Windows AD ✓
Repo	github.com/osim-framework/osim-core	Firewall	T620 OPNsense 26.1.5	Switches	Cisco 3560CX + SF300-24P

1. Executive Summary

This report documents the successful completion of Phase 1 (Network Architecture) and Phase 2 (Windows Enterprise Target Environment) of the OSIM Lab build. The lab now has a fully operational blue/red separated network with 12 VLANs, a production-grade firewall ruleset, and a complete Windows Server 2022 Active Directory domain with two workstations. All four Wazuh agents are active and generating telemetry. Sysmon with the Olaf Hartong modular configuration is deployed on all Windows hosts.

2. Phase 1 — Network Architecture

2.1 VLAN Architecture

VL AN	Name	Subnet	Gateway	Purpose	Status
10	MGMT-A	10.10.10.0/24	10.10.10.1	Proxmox/infra admin	Existing ✓
11	MGMT-B	10.10.11.0/24	10.10.11.1	MacBook management	Existing ✓
20	LAB-OPS	10.10.20.0/24	10.10.20.1	TheHive, MISP	Existing ✓
30	WINDOWS-ENT	10.10.30.0/24	10.10.30.1	AD DC, workstations	NEW ✓
40	SECURITY	10.10.40.0/24	10.10.40.1	Wazuh, Velociraptor, Caldera	Existing ✓
50	RED-OPS	10.10.50.0/24	10.10.50.1	T480 attacker tooling	NEW ✓
60	C2-ISOLATED	10.10.60.0/24	10.10.60.1	Sliver/Havoc C2 listeners	NEW ✓
70	MAL-SANDBOX	10.10.70.0/24	10.10.70.1	Malware detonation — no WAN	NEW ✓
80	FAKE-INTERNET	10.10.80.0/24	10.10.80.1	INetSim fake internet	NEW ✓
90	OSWE-WEB	10.10.90.0/24	10.10.90.1	Web app targets for OSWE	NEW ✓
91	OSED-TARGET	10.10.91.0/24	10.10.91.1	Exploit dev Windows targets	NEW ✓

92	JUMP-ADMIN	10.10.92.0/24	10.10.92.1	Hardened jump host	NEW ✓
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2.2 Cisco Switch Configuration

Port	Name	Mode	VLAN(s)	Connected To
Gi0/2	T620-MGMT-B	Access	11	OPNsense igb0 (MGMTB)
Gi0/3	T620-MGMT-A	Access	10	OPNsense igb1 (MGMTA)
Gi0/4	T620-LAB	Trunk	20,30,50,60,70,80,90,91,92 native=20	OPNsense igb2 (LAB+new VLANs)
Gi0/5	T620-SECURITY	Access	40	OPNsense igb3 (SECURITY)
Gi0/6	P520-PROXMOX	Trunk	11,20,30,40,50,60,70,80,90,91,92	P520 Proxmox vmbr0
Gi0/7	MACBOOK-MGMT B	Access	11	MacBook management

2.3 Key Firewall Rules (OPNsense)

Rule ID	Interface	Source	Destination	Port	Action
WIN-01-03	WINDOWSENT	WINDOWSENT net	SECURITY net	1514,1515,55000	PASS
WIN-04-05	WINDOWSENT	WINDOWSENT net	SECURITY net	8000,8001,8888	PASS
WIN-07-08	WINDOWSENT	WINDOWSENT net	C2ISOLATED net	80,443	PASS
RED-01	REDOPS	REDOPS net	WINDOWSENT net	Any	PASS
RED-05	REDOPS	REDOPS net	SECURITY net	Any	BLOCK
RED-06	REDOPS	REDOPS net	LAB net	Any	BLOCK
C2-01	C2ISOLATED	C2ISOLATED net	FAKEINTERNET net	Any	PASS
C2-02	C2ISOLATED	C2ISOLATED net	WAN	Any	BLOCK
MAL-01	MALSANDBOX	MALSANDBOX net	FAKEINTERNET net	Any	PASS
MAL-02	MALSANDBOX	MALSANDBOX net	WAN	Any	BLOCK
SEC-01	SECURITY	SECURITY net	WINDOWSENT net	Any	PASS

SEC-02	SECURITY	SECURITY net	Any	Any	PASS
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2.4 Phase 1 Validation

Test	Command	Result
All 8 new gateway IPs	ping -c 2 10.10.X0.1 (OPNsense)	PASS — all 8 respond
VLAN 30 trunk (Gi0/4)	show interfaces gi0/4 trunk	PASS — 20,30,50-92 forwarding
ARP VLAN 30	tcpdump -ni vlan01 arp	PASS — DC01 ARP resolves
Block red→blue	pfctl -sr grep vlan02.*igb3	PASS — block rule confirmed
Block sandbox→WAN	pfctl -sr grep vlan04.*re0	PASS — block rule confirmed

3. Phase 2 — Windows Enterprise Target Environment

3.1 VM Inventory

VMI D	Name	OS	RAM	IP	VLA N	Role	Status
100	wazuh-manage r	Ubuntu 22.04	16GB	10.10.40.10	40	Wazuh SIEM	Active ✓
101	thehive	Ubuntu 22.04	4GB	10.10.20.10	20	Case mgmt	Active ✓
102	misp	Ubuntu 22.04	4GB	10.10.20.11	20	Threat intel	Active ✓
106	win-dc01	WS 2022	6GB	10.10.30.10	30	AD DC, DNS, osim.lab	Active ✓
107	win-client01	Windows 11	4GB	10.10.30.20	30	Domain workstation	Active ✓
108	win-client02	Windows 10	4GB	10.10.30.21	30	Domain workstation	Active ✓

3.2 Active Directory Configuration

Component	Value
Domain Name	osim.lab
NetBIOS Name	OSIM
Forest/Domain Mode	Windows Server 2016 (WinThreshold)
Domain Controller	DC01.osim.lab — 10.10.30.10
FSMO Roles	All five roles held by DC01 (PDCEmulator, RIDMaster, InfrastructureMaster, SchemaMaster, DomainNamingMaster)
DNS	Integrated DNS on DC01, forwarder: 10.10.11.1 (OPNsense)
OUs Created	OSIM-Users, OSIM-Admins, OSIM-Workstations, OSIM-Servers, OSIM-ServiceAccounts
User Accounts	jsmith, sjones, mchen, lbrown, tdavis (OSIM-Users) helpdesk (OSIM-Admins, Domain Admin) svc-backup, svc-sql (OSIM-ServiceAccounts)
Workstations	CLIENT01 (Win11, 10.10.30.20), CLIENT02 (Win10, 10.10.30.21) — both domain joined

3.3 Detection Stack Deployment

Host	Sysmon	Config	Wazuh Agent	Agent Status
DC01	v15.90	Olaf Hartong modular (sysmonconfig.xml)	Enrolled → 10.10.40.10	Active ✓
CLIENT01	v15.90	Olaf Hartong modular (sysmonconfig.xml)	Enrolled → 10.10.40.10	Active ✓

CLIENT02	v15.90	Olaf Hartong modular (sysmonconfig.xml)	Enrolled → 10.10.40.10	Active ✓
OPNsense	N/A	Native OPNsense Wazuh agent	Enrolled → 10.10.40.10	Active ✓

3.4 Wazuh Dashboard — 24hr Alert Summary

Severity	Rule Level	Count	Source
Critical	Level 15+	0	No critical alerts at baseline
High	Level 12-14	0	No high alerts at baseline
Medium	Level 7-11	285	Sysmon process creation, network connections, registry changes on Windows hosts
Low	Level 0-6	5,017	Normal AD authentication, DNS queries, service starts, Sysmon file events

3.5 Snapshots Taken

VMI D	VM Name	Snapshot Name	Description
106	win-dc01	dc01-ad-baseline	DC01 osim.lab domain promoted, OUs and users created, RDP enabled
106	win-dc01	dc01-wazuh-sysmon	DC01 Sysmon v15.90 + Wazuh agent active, osim.lab domain live
107	win-client01	win11-domain-joined	CLIENT01 Win11 domain joined osim.lab, Sysmon + Wazuh active
108	win-client02	win10-domain-joined	CLIENT02 Win10 domain joined osim.lab, Sysmon + Wazuh active

4. Phase 2 Validation Gates

Gate	Test	Command / Method	Result
2.1	DC01 network connectivity	nc -zv 10.10.30.10 3389 (from OPNsense)	PASS — Connection succeeded
2.2	osim.lab domain promoted	Get-ADDomain (on DC01)	PASS — DNSRoot: osim.lab, PDC: DC01.osim.lab
2.3	Sysmon running on DC01	.\Sysmon64.exe -accepteula -i sysmonconfig.xml	PASS — Sysmon64 started, SysmonDrv started
2.4	Wazuh agent DC01 active	Wazuh dashboard → Agents → Summary	PASS — DC01 agent Active
2.5	CLIENT01 domain joined	(Get-WmiObject Win32_ComputerSystem).Domain	PASS — returns osim.lab
2.6	CLIENT02 domain joined	(Get-WmiObject Win32_ComputerSystem).Domain	PASS — returns osim.lab
2.7	4 Wazuh agents active	Wazuh dashboard → Agents → Summary	PASS — Active(4): OPNsense, DC01, CLIENT01, CLIENT02
2.8	Telemetry generating	Wazuh dashboard → Last 24hr alerts	PASS — 285 medium + 5,017 low alerts confirmed

5. Pending — Next Phases

Phase	Component	VMID	Description	Priority
3	Greenbone	103	Vulnerability scanner — VLAN 40, scans WINDOWSENT	HIGH
3	Velociraptor	104	Live response server — VLAN 40, agents on all Windows hosts	HIGH
3	Caldera	105	Adversary simulation server — VLAN 40, ATT&CK; operations	HIGH
3	win-server01	109	Member server — IIS, MSSQL, SMB, WinRM on VLAN 30	MEDIUM
3	jump-admin	110	Debian jump host — RSAT, RDP client on VLAN 92	MEDIUM
4	T480 red node	200+	Kali Purple, Sliver C2, REMnux, CAPE, INetSim on T480	HIGH
4	Atomic Red Team	—	First ATT&CK; simulation run, verify Wazuh→TheHive pipeline	HIGH
5	OpenCTI	—	Threat intel platform — GCTI cert prep	LOW
5	SOAR automation	—	n8n or Shuffle — Wazuh→TheHive→MISP automated workflow	LOW

6. Known Issues and Recommendations

ID	Issue	Impact	Recommendation
KI-01	T620 igb2/igb3 cable flapping	Intermittent no carrier on LAB and SECURITY interfaces. Required physical reseal and switch port bounce multiple times during build.	Replace or reseal cables. Label all patch cables. Consider replacing with higher quality Cat6 cables.
KI-02	P520 bridge-vids 2-40 too broad	bridge-vids 2-40 adds all VLANs 2-40 to nic0. Only VLANs 20, 30, 40 are needed. Creates unnecessary VLAN entries.	Update /etc/network/interfaces to use bridge-vids 1 20 30 40 instead of 2-40 after validating all VMs work correctly.
KI-03	Wazuh dashboard HTTPS rule ordering	MB-WAZUH-DASHBOARD rule was initially placed below MB-99 TERMINATOR, blocking access from MacBook.	Rule has been fixed. When adding new MGMTB rules always ensure they are placed above the MB-99 TERMINATOR.
KI-04	Windows VMs no internet access	WINDOWSENT VLAN has no WAN egress rule. Software downloads require ISO upload via MacBook→Proxmox SCP workflow.	Add temporary internet egress rule when needed for downloads. Remove after use to maintain isolation model.
KI-05	Wazuh agent 8.8.8.8 DNS on Wazuh VM	Wazuh VM netplan still has 8.8.8.8 as secondary DNS. DNS-BLOCK floating rule on OPNsense will drop these queries.	Remove 8.8.8.8 from /etc/netplan/50-cloud-init.yaml on VM100. Use only 10.10.11.1.